# MD. Mehedi Hasan Maruf

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## Education

### B.Sc. Engineering in CSE *Graduated: 08/2026*

*Rajshahi University of Engineering & Technology (RUET)* | CGPA: 3.75 / 4.00, Honours

## Skills

**AI/ML:** PyTorch (GNNs), Transformers, LangChain, RAG, NLP, causal inference (DoWhy), Bayesian
**Languages:** Python, C++, TypeScript, JavaScript, SQL
**CS Fundamentals:** Data Structures & Algorithms, Graph Algorithms, Number Theory, Software Design
**Systems & Tools:** Linux, Docker, CI/CD, FastAPI, Django, Next.js (React), PostgreSQL

## Experience

### Software Engineer II *Jan 2025 – Present*

*Aamira Software Solutions*, Kokomo, Indiana, USA (Remote)

- Raised client response rates by 30% by leading development of a CRM system.
- Cut API response time by 40% by architecting and implementing a TypeScript microservice backend.
- Improved application performance by 60% by profiling and optimizing PostgreSQL/Supabase database queries.

### Junior Software Engineer *Nov 2022 – Apr 2024*

*Code Studio*, Rajshahi, Bangladesh

- Kept 12 Linux servers at 99.8% uptime; cut deployment time by 40% with CI/CD and Docker across 5 client setups.
- Built Python (Django) and JavaScript/TypeScript backends supporting thousands of daily users.

## Research

### TRACE: Curriculum Bottleneck Detection & Causal Validation *Jul 2025 – Aug 2026*

*B.Sc. Thesis, Dept. of CSE, RUET*

- Built a Python framework that ranks the courses holding students back, from prerequisite-graph position (NetworkX, GAT in PyTorch), Bayesian fail rates, and feedback sentiment (DeBERTa/VADER).
- Reached F-measure 0.75 and NDCG@10 0.81 on a reproducible synthetic benchmark with planted bottlenecks.
- DoWhy causal check on 23,369 OULAD students: 19.9-point effect of high assignment engagement on pass rates ($p < 0.01$).
- Under review at LAK 2027: *When Agreement Misleads: Auditing Course Shortlists in Curriculum Analytics.*

## Selected Projects

### Omnious | Visual Debugging Platform *Nov 2025 – Present*

- Building an AI-assisted debugger that draws a program's code graph (TypeScript, Fastify, tRPC, Next.js).
- Semantic code search with PostgreSQL pgvector, ltree, and pg_trgm under Row-Level Security; Docker, GitHub Actions.

### StudentAI | Intelligent Learning Companion *Mar 2025*

- Built a RAG learning assistant (Python/FastAPI, LangChain, Next.js) with semantic retrieval, conversation memory, multimodal ingestion (PDF, image, video), and quiz generation.

## Certifications & Achievements

- 4th Place, RUET CodeSmash (team, 2023) and CodeSpark (individual, 2022) programming contests; Competitive Programming Mentor, RAPL, RUET (2023), teaching number theory and graph algorithms.